

Use-Case Flow Specification

Use Case: Complete Purchase with Inspection and Escrow

Primary Actor: Buyer **Supporting Actors:** Inspection Agent, Payment Gateway

Goal: Complete a second-hand product purchase while ensuring payment remains in escrow until inspection and buyer sign-off are completed.

Preconditions

- Buyer is authenticated and has selected an available product.
- Product is eligible for inspection and payment service is available.
- Inspection agent is available.

Postconditions

- Success: inspection is recorded, buyer signs off, and escrow funds are released.
- Alternate: disputed escrow remains frozen while the dispute is active.

Main Success Scenario

1. Buyer selects an available product and chooses Purchase.
2. System creates the order and requests payment details.
3. Buyer submits valid payment details.
4. Payment Gateway authorizes the payment.
5. System locks the authorized amount in escrow and changes the order state to Under Inspection.
6. System creates an inspection request and assigns it to an Inspection Agent.
7. Inspection Agent examines the product and submits the physical verification result.
8. System stores the inspection result and notifies the Buyer.
9. Buyer reviews the inspection result and provides explicit sign-off.
10. System changes the escrow status to Released and releases the payment to the seller.
11. System marks the transaction as completed and confirms completion to the Buyer.

Alternate Flow – Buyer Raises a Dispute

- 9a1. At Step 9, Buyer rejects the inspection result and selects Raise Dispute.
- 9a2. System creates a dispute ticket and immediately freezes the related escrow funds.
- 9a3. System blocks withdrawal/release of the disputed amount while the dispute is active and notifies relevant parties.
- 9a4. Use case ends in Dispute Pending; escrow is not released.